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Document details

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Purpose	Trace IOB System HW & SW evolution
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Document control

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Contents

1. Introduction

1.1 Document Purpose

The purpose of this document is to describe the constituent components of the project software build developed by Nomad for deployment to **ÖBB DOSTO Neu** project's Internet-on-Board (IoB) solution.

As part of Nomad's quality assurance and change management processes, this document also provides a summary of the change log(s) and release history associated with the software release(s) through its evolution, with references made (where applicable) to Test Exit Report(s).

This covers the hardware and software deployed on the trains across all four DOSTO Neu platform variants: nv4 (4734, 4-car), nv6 (4736, 6-car), fv5 (4705, CAT 5-car) and fv6 (4706, FV 6-car).

The following documents give more details about the project:

Title	Reference	Short Summary
System Design Document (SDD)	ND-DEL-OBB-035-SDD-002-01	Onboard network system design for the DOSTO Neu IoB solution
Nomad Connect DEVOPS TER report	ND-DEVOPS-OBB-TER-026-04	Nomad Connect DEVOPS TEST EXIT REPORT

Maturity Level

All current versions of software described in this version of the document are delivered for:

		Comments
	Tests on train	
X	Pre-commercial service (deployment on delivered train)	v8 rollout in progress across the DOSTO Neu fleet (nv4, nv6, fv5, fv6)
	Customer Test on train	
	Commercial service (fleet deployment)	

2. Hardware Platform Compatible With Delivered Software Release

This chapter mentions the current HW versions / Firmware of all the equipment part of **ÖBB DOSTO Neu** System when applicable compatible with delivered software release described in §3 of this document.
Each time HW Version / Firmware Version will change, the current versions will be updated in the "current version" paragraph and content of the evolutions will be clearly explained in the "HW Version evolution".

List of Hardware

NMID	Equipment description	Supplier	Comments
1138	Nomad Communication & Control Unit (CCU) – R5001C	Nomad Digital	1 per trainset; onboard server / cellular gateway
1143	VDS Rail Consist Switch – VDSRail DS-CSWP28 (28-port, 10 GbE)	Nomad Digital	3 per car (12 nv4 / 18 nv6 / 15 fv5 / 18 fv6)
1043	Westermo Ibex-610 Wi-Fi access point (RT610LV / IbexOS)	Nomad Digital	4 per car (16 nv4 / 24 nv6 / 20 fv5 / 24 fv6); passenger Wi-Fi
1109	5G modem – Telit FN990A28 (LM990 carrier), integrated in CCU	Nomad Digital	Aggregated on CCU bond0 for WAN backhaul
720	GNSS / GPS receiver – uBlox NEO-M8L (integrated in CCU)	Nomad Digital	Passive – no firmware (NA)
N/A	Roof / Wi-Fi antennas – Sencity Rail Multi (1399.99.0307), Huber&Suhner indoor (1399.35.0020)	Stadler	Passive – no firmware (NA)

The onboard network hardware above is common across all four DOSTO Neu platform variants; the switch and access-point counts and the switch configuration template differ per variant as follows:

Platform	Series	Cars	Switches	Access Points	Configuration template
nv4	4734	4	12	16	nd-obn-template-dostoneu-nv4
nv6	4736	6	18	24	nd-obn-template-dostoneu-nv6
fv5 (CAT)	4705	5	15	20	nd-obn-template-dostoneu-fv5
fv6 (FV)	4706	6	18	24	nd-obn-template-dostoneu-fv6

System Description

2.1 Onboard Network Hardware (CCU, VDS Consist Switch, Westermo AP)

2.1.1.1 Current Version

The current firmware and configuration baseline of the onboard network hardware, common across all four DOSTO Neu platform variants (nv4, nv6, fv5, fv6), is:

- VDS Rail Consist Switch – firmware 7.4.2; configuration V8 (per-platform template – nv4 / nv6 / fv5 / fv6 – with the per-train Fzg-ID rendered into switch hostnames).
- Westermo RT610LV Wi-Fi access point – firmware 6.11.2-0 (IbexOS); Nomad access-point configuration.
- Nomad CCU – Nomad Connect platform 2025.2.1 (validated build; configuration ID 7ce1d29ce6). Onboard-network engine (nd-obn) 2.2.23.

2.1.1.2 HW Version Evolution

Hardware versions are read from the equipment labels.

3. Software Packages Delivered

This chapter mentions the current software release versions part of **ÖBB DOSTO Neu** delivery when delivered with compatible hardware as described in §2 of this document.

Each time software release will change, the current versions will be updated in the “current version” paragraph and content of the evolutions will be clearly explained in the **“SW Version Evolution”.**

Delivered Software Table		
SW#	**SW Version#**	**Description**
SW1	*7.4.2*	*VDS Rail Consist Switch firmware*
SW2	*V8*	*VDS Rail Consist Switch configuration (per-platform template – nv4 / nv6 / fv5 / fv6; per-train Fzg-ID)*
SW3	*6.11.2-0*	*Westermo RT610LV Wi-Fi access point firmware (IbexOS)*
SW4	*2.2.23*	*Nomad Connect / onboard-network engine (nd-obn) on the CCU*
SW5	*2025.2.1*	*CCU platform image (Nomad Connect base software)*

Software Testing Tools

Delivered Software Table		
Package Name	**Version**	**Description**
Nomad NMS (Zabbix-based)	*2026.1.2*	*Fleet network monitoring / SNMP polling used to verify device reachability and status*

4. Characteristics

New Functionalities

This is the first issue of the release note. It records the baseline onboard-network software and firmware deployed to the DOSTO Neu fleet: VDS Consist Switch configuration V8 with firmware 7.4.2, Westermo access-point firmware 6.11.2-0, and Nomad Connect (nd-obn) 2.2.23 on the CCU. No new functionality is introduced relative to a previous release note; subsequent updates will list added functions here with their reference (ticket / headline).

Omissions and Restrictions

- Some Westermo access points may report firmware as staged but not activated after an update; these are re-triggered on the next commissioning visit. Affected units are tracked per train in the fleet status records.
- Stadler-side firewall commissioning and a small number of open cabling faults are outside Nomad’s scope and are tracked separately; they do not affect the Nomad onboard-network baseline recorded here.

No other functional restrictions apply to the onboard-network baseline recorded in this release.

Media Content

Not applicable. No portal media content is delivered by Nomad as part of this release.

Testing and Validation

The onboard-network baseline is validated per train through a Layer-2 network health check (switch error counters, RSTP topology, trunk states, and end-to-end vlan7 reachability to the Stadler firewall) and, where applicable, throughput sampling. Results are recorded per train and summarised in the customer network health-check reports. Software validation is covered by the Nomad Connect DEVOPS Test Exit Report referenced in Section 1.